12TH BIOLOGY · CHAPTER 13 · FULL CHAPTER NOTES

Thermoregulation and Osmoregulation

Topic-wise short and long questions with answers from the complete chapter

13.1 HOMEOSTASIS

Q.1 [Short] What is homeostasis? Which important processes are included in it?

Homeostasis is a vital process by which living organisms maintain a stable internal environment, despite changes in their external environment. It consists of a wide range of mechanisms that work together to regulate the internal environment of the body. Its important processes are **thermoregulation**, **osmoregulation** and **excretion**.

13.1.1 Basic Mechanism of Homeostasis

Q.2 [Long] How do receptors, control centre and effectors work together to maintain homeostasis?

Homeostasis is maintained through the collaboration of three essential elements.

1. Receptors

Receptors or sensors are specialized structures (organs, tissues, cells or subcellular components) that detect changes in the internal or external environment. They transmit the information of the change to the control centre. For example, receptor cells in the skin detect changes in temperature, and receptor cells in the blood vessels detect changes in blood pressure.

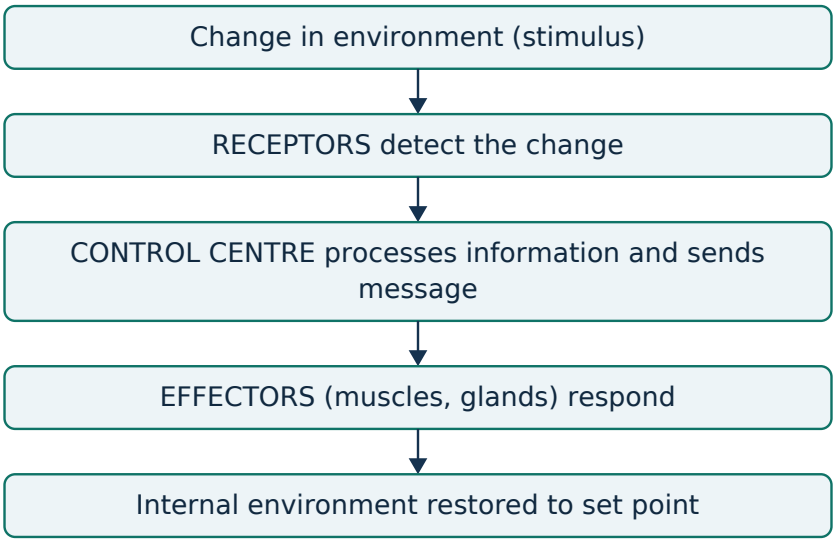
2. Control centre

The control centre receives information from the receptors, processes it and sends an appropriate message to the effectors. The brain, spinal cord and endocrine glands act as control centres.

3. Effectors

Effectors are the parts of the body which respond when they receive a message from the control centre. Muscles and glands are effectors. Muscles respond by contracting and glands respond by releasing their secretions. These responses restore the internal environment to its set point.

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Elements of homeostasis

Q.3 [Short] What are set points? How do control centres help to maintain homeostasis?

Set points are the values of the internal environment (e.g., body temperature of 37 °C) which are optimal for the functioning of the body. Control centres receive information from the receptors, process it and send messages to the effectors. These messages make sure that the internal environment remains within a narrow range of the set points.

Q.4 [Short] How does the body respond when its temperature drops below the set point?

When the body temperature drops below the set point, the effectors in the skin and muscles cause **shivering** and **vasoconstriction**. Shivering increases heat production and vasoconstriction reduces heat loss. As a result, the body temperature rises back to the set point.

Q.5 [Short] Which structures act as receptors, control centres and effectors in the human body?

Element	Structures	Function
Receptors	Receptor cells in skin, blood vessels, hypothalamus	Detect changes and inform the control centre
Control centre	Brain, spinal cord, endocrine glands	Process information and send messages to effectors
Effectors	Muscles and glands	Muscles contract; glands release secretions

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13.1.2 Feedback Mechanisms

Q.6 [Short] Why are feedback mechanisms called the foundations of homeostasis?

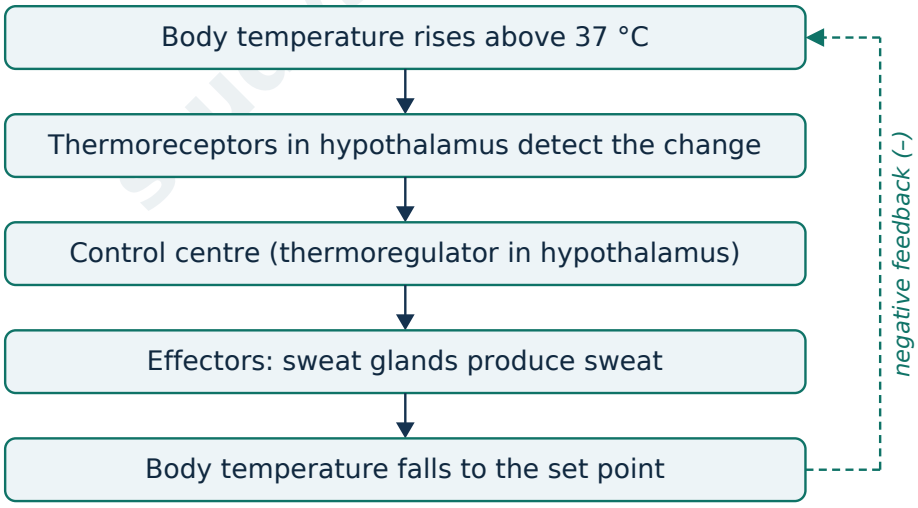
Feedback mechanisms are the regulatory processes in which the output or product of a system is used to control the behaviour of the system itself. Through feedback, the body "knows" the result of its own response and can reduce or increase that response. Without feedback, the internal environment could not be kept near the set points. There are two types: negative feedback and positive feedback.

Q.7 [Long] Explain the negative feedback mechanism with the example of body temperature.

A negative feedback system is a mechanism in which the output or product of a system **reduces** the activity of the system. It is the most common form of regulation. Common examples are the regulation of body temperature, blood glucose level and hormone secretion.

Example: control of body temperature

1. The body temperature rises above the set point of 37 °C.
2. Thermoreceptors in the hypothalamus of the brain detect this change.
3. They send signals to the control centre (thermoregulator in the hypothalamus).
4. The control centre sends messages to the effectors, e.g., sweat glands.
5. The sweat glands produce sweat, which lowers the temperature.
6. The output (temperature back at the set point) influences the thermoreceptors negatively, so they stop sending messages to the control centre.



Negative feedback in the control of body temperature

Q.8 [Long] Explain the positive feedback mechanism with the example of childbirth.

A positive feedback system is a regulatory mechanism in which the output or product **increases** the activity of the system. In this way, positive feedback enhances the original stimulus. It often happens in processes that require rapid and significant changes, e.g., blood clotting, labour and childbirth.

Example: childbirth

1. Contractions of the uterus send nerve impulses to the control centre (a part of the hypothalamus).
2. The control centre sends signals to the pituitary gland to release oxytocin hormone.
3. Oxytocin stimulates further contractions of the uterus.
4. Stronger contractions send more nerve impulses to the hypothalamus, so more oxytocin is released.
5. This positive feedback loop continues until the baby is born.

Q.9 [Short] Why is positive feedback less common in the body than negative feedback?

Homeostasis means keeping the internal environment stable near the set points. Negative feedback reverses a change and brings the body back to the set point, so it is used most commonly. Positive feedback enhances the original stimulus and takes the body further away from the starting condition. So it is useful only in the few processes that need a rapid and significant change, e.g., blood clotting and childbirth.

Q.10 [Short] How can unregulated positive feedback cause health problems?

In severe inflammation or infection, the immune cells release proteins called **cytokines**. These proteins further stimulate the immune response, which releases more cytokines. If this positive feedback is not properly regulated, it leads to excessive inflammation and tissue damage.

Q.11 [Short] How does negative feedback differ from positive feedback?

Negative feedback	Positive feedback
The output of the system reduces the activity of the system.	The output of the system increases the activity of the system.
It reverses the change and restores the set point.	It enhances the original stimulus.
It is the most common form of regulation.	It is less common.
Examples: regulation of body temperature, blood glucose level, hormone secretion.	Examples: blood clotting, labour and childbirth (oxytocin).

13.2 OSMOREGULATION

Q.12 [Short] What is osmoregulation? Why is it essential for the body?

The maintenance of the balance of water and solutes in the body fluids is called osmoregulation. It ensures that the body fluids do not become too dilute or too concentrated. It is essential for many physiological processes, including maintaining proper blood pressure, enabling the proper function of organs and regulating metabolism.

Q.13 [Short] What is osmosis? What is meant by the osmotic pressure of a solution?

Osmosis is the diffusion of water across a membrane. It always occurs from a more dilute (hypotonic) solution to a less dilute (hypertonic) solution. The osmotic pressure of a solution is a measure of its tendency to take in water by osmosis.

13.2.1 Osmoregulators and Osmoconformers

Q.14 [Short] Why are osmoconformers usually found in marine water?

Osmoconformers cannot maintain the concentration of water and solutes in their bodies. The concentration of their body fluids changes with the surrounding medium and they remain isotonic to it. So they can live safely only in environments where the solute concentration is relatively stable, such as marine water. Examples are all marine invertebrates, some freshwater invertebrates and some marine vertebrates like hagfishes, sharks and rays.

Q.15 [Short] How do osmoregulators keep their internal concentration constant?

Osmoregulators can regulate the concentrations of water and solutes in their body fluids. They keep a constant internal concentration even when the concentration of the environment changes. For this purpose they use mechanisms such as active transport of salts, accumulation of salts and water conservation. Most vertebrates of aquatic and terrestrial environments are osmoregulators.

Q.16 [Short] How do osmoconformers differ from osmoregulators?

Osmoconformers	Osmoregulators
They allow the osmotic concentration of their body fluids to match the environment.	They regulate the concentrations of water and solutes in their body fluids.
They remain isotonic to their environment.	Their internal concentration is constant and may differ from the environment.
They live in environments with stable solute concentration, e.g., marine water.	They can live in changing environments, both aquatic and terrestrial.
Examples: all marine invertebrates, hagfishes, sharks, rays.	Examples: most vertebrates.

13.2.2 Problems and Methods of Osmoregulation

Q.17 [Short] What osmotic problems are faced by freshwater animals?

Freshwater animals are generally **hypertonic** to their environment, i.e., their body fluids have a higher solute concentration than the surrounding water. So they face two problems:

- Water continuously enters their bodies through osmosis, particularly across permeable surfaces like gills.

- Essential salts and ions (like Na^+ and Cl^-) diffuse out of the body into the surrounding water.

Q.18 [Short] How do freshwater animals solve their osmotic problems?

- They prevent water from entering their bodies and eliminate the excess water by producing **large volumes of dilute urine**.
- They **actively transport salts** back into their bodies to compensate for the loss of salts.
- Unlike marine animals, they **rarely drink water**, because water is already entering by osmosis.

Q.19 [Short] What osmotic problems are faced by marine animals?

Most marine animals are **hypotonic** to their environment, i.e., seawater has a much higher solute (salt) concentration than their internal tissues. So:

- Water constantly leaves their bodies by osmosis through the gills and skin.
- Salts continuously diffuse from the environment into their bodies.
- They also get an excess of salts in the body due to drinking of seawater.

Q.20 [Short] Marine fishes drink seawater continuously. How do they get rid of the excess salts?

- **Chloride cells:** Specialized chloride cells in the gills use energy (ATP) to actively pump salts out of the blood and back into the environment.
- **Kidneys:** They also remove excess salts through their kidneys, in a very small amount of highly concentrated urine.
- **Rectal glands:** Some marine fishes have special salt-excreting glands in the rectum. These glands remove salts from the blood into the digestive tract, and the salts are eliminated along with the faeces.

Q.21 [Long] Compare osmoregulation in freshwater animals and marine animals.

Feature	Freshwater animals	Marine animals
Body fluids	Hypertonic to the environment	Hypotonic to the environment
Movement of water	Water enters the body by osmosis (mainly through gills)	Water leaves the body by osmosis (through gills and skin)
Movement of salts	Salts (Na^+, Cl^-) diffuse out of the body	Salts diffuse into the body
Drinking	Rarely drink water	Drink seawater continuously
Urine	Large volume of dilute urine	Very small amount of highly concentrated urine
Salts	Salts are actively transported into the body	Salts are actively pumped out by chloride cells of gills; also removed by kidneys and rectal glands

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Q.22 [Long] How do terrestrial animals maintain their water balance?

Terrestrial animals lose water to the environment through evaporation from respiratory surfaces, through the skin and via the excretion of wastes. They have to make sure that water intake (from drinking and food) plus metabolic water production is equal to the water lost. They have anatomical, physiological and behavioural adaptations for this purpose.

- **Impermeable outer layers (anatomical):** These prevent evaporative water loss. Insects possess a waxy exoskeleton made of chitin. Reptiles have waterproof scales made of keratin. Mammals and birds have dead, keratinized skin cells and oil-secreting glands in the skin.
- **Kidneys (physiological):** Their kidneys play a vital role in retaining water inside the body.
- **Less toxic wastes (physiological):** They convert ammonia, the primary waste product of protein metabolism, into urea or uric acid, which require less water for excretion.
- **Behaviour:** Many desert animals stay underground during the heat of the day to avoid evaporation. They emerge only at night, when the air is cooler and more humid.
- **Metabolic water:** Desert animals feed upon seeds of desert plants, which store large amounts of carbohydrates. Some animals, like the kangaroo rat, survive entirely on metabolic water.

Q.23 [Short] What is metabolic water? How does the kangaroo rat survive without drinking water?

Metabolic water is the water produced as a by-product of cellular respiration. The kangaroo rat feeds on seeds of desert plants, which are rich in carbohydrates. When these carbohydrates are broken down in cellular respiration, water is produced. The kangaroo rat survives entirely on this metabolic water.

Q.24 [Short] What is anhydrobiosis?

Anhydrobiosis is a dormant state shown by certain invertebrates, like tardigrades (water bears). They enter this state when they lose almost all their body water. They come back to active life, even years later, when water returns.

13.2.3 Nature of Excretory Products in relation to Habitat

Q.25 [Short] How is ammonia produced in the body? Why must it be excreted rapidly?

Ammonia (NH_3) is produced in the liver when the amino group ($–NH_2$) is removed from amino acids and nucleic acids and is combined with H^+. Ammonia is highly toxic, so it cannot be stored in the body and is excreted rapidly.

Q.26 [Short] Why can only aquatic animals afford to excrete ammonia?

One gram of ammonia requires 500 mL of water to dissolve it to a non-toxic level. Such plenty of water is available only to animals living in water, e.g., most fishes, protozoans, sponges and coelenterates. Marine animals excrete ammonia directly into the water. Animals which excrete ammonia as their major nitrogenous waste are called **ammonotelic**.

Q.27 [Short] Why do terrestrial animals convert ammonia into urea?

Terrestrial animals face the challenge of conserving water, so they cannot afford to excrete ammonia with plenty of water. They convert ammonia into urea, $CO(NH_2)_2$, in the liver and excrete it through the kidneys. Urea is less toxic than ammonia and one gram of urea requires only 50 mL of water. Such animals are called **ureotelic**.

Q.28 [Short] Birds, reptiles and insects spend a lot of energy to make uric acid. What is the benefit?

These animals live in arid environments and face an extreme water conservation challenge. They cannot afford to excrete even urea. Uric acid ($C_5H_4N_4O_3$) is even less toxic than urea, is less soluble in water and makes precipitates. One gram of uric acid requires just 1 mL of water for its excretion. So the energy spent is repaid by a great saving of water. Such animals are called **uricotelic**.

Q.29 [Long] Relate the three nitrogenous wastes with the habitat of animals.

Feature	Ammonia	Urea	Uric acid
Formula	NH_3	$CO(NH_2)_2$	$C_5H_4N_4O_3$
Toxicity	Highly toxic	Less toxic than ammonia	Even less toxic than urea
Water needed for 1 g	500 mL	50 mL	1 mL
Solubility	Dissolves in plenty of water	Dissolves in water	Less soluble; makes precipitates
Habitat	Plenty of water available	Land (water must be conserved)	Arid environment (extreme water conservation)
Animals	Most fishes, protozoans, sponges, coelenterates	Terrestrial animals, e.g., humans	Insects, desert animals, birds, most reptiles
Name of group	Ammonotelic	Ureotelic	Uricotelic

So, as the availability of water decreases from aquatic to terrestrial to arid habitats, animals excrete a less toxic waste that needs less water.

Q.30 [Short] Why do humans excrete uric acid even though they are not uricotelic? What is gout?

Humans are ureotelic, but they also excrete a small amount of uric acid in urine. This uric acid is a by-product of the breakdown of **purines**, which are found in many foods, particularly meat and some types of vegetables. It is produced in the liver. In healthy individuals, most of it is excreted in the urine. When there is an excess of uric acid in the blood, uric acid crystals are formed in joints and tissues. This painful condition is called **gout**.

13.3 THERMOREGULATION

Q.31 [Short] What is thermoregulation? Why is it so important for animals?

Thermoregulation is the homeostatic process by which organisms maintain their internal body temperature within a tolerable range, regardless of the temperature of the external environment. It is important for two reasons:

- **Enzymes:** Enzymes have an optimal temperature at which they function most efficiently. If the temperature falls below the optimal, enzyme–substrate collisions decrease and metabolic rates drop. If it rises well above the optimal, enzymes begin to denature and lose their function permanently.
- **Cell membranes:** The lipid bilayer of cell membranes is temperature-sensitive. At high temperatures, cell membranes can lose their selective permeability. Low temperatures can make membranes rigid and brittle.

13.3.1 Types of Animals on the basis of Heat Source

Q.32 [Short] Why can ectotherms not maintain their body temperature by themselves?

In ectotherms, the rate of metabolism is very low. They cannot produce enough heat to raise and maintain their body temperature. So they rely on heat from the environment, and their internal temperature generally fluctuates with the temperature of the surroundings. Most invertebrates, fishes, amphibians and reptiles are ectotherms.

Q.33 [Short] What is behavioural thermoregulation? How do ectotherms use it?

Behavioural thermoregulation is the control of body temperature by changing behaviour or position instead of producing heat inside the body. Ectotherms use it in the following ways:

- If they are too cold, they bask in the sun (heat gain by radiation) or sit on a warm rock (heat gain by conduction).
- If they are too hot, they seek shade or burrow into the cool earth.
- In very cold climates, they become dormant during the coldest months.

Q.34 [Short] How do endotherms produce heat and how do they retain it?

Endotherms (birds and mammals) generate the majority of their body heat through internal metabolism. They maintain a relatively constant body temperature, often higher than the environment. When it gets cold, they may shiver (muscle contractions) or burn brown adipose tissue for heat. To retain this heat, most endotherms have insulation such as hair, feathers or fat.

Q.35 [Short] How do ectotherms differ from endotherms?

Ectotherms	Endotherms
Gain most of their body heat from external sources.	Generate the majority of their body heat through internal metabolism.
Metabolic rate is very low.	Metabolism produces enough heat.
Body temperature fluctuates with the surroundings.	Body temperature is relatively constant, often higher than the environment.
Use behavioural thermoregulation (basking, shade, burrowing, dormancy).	Shiver, burn brown adipose tissue; have insulation (hair, feathers, fat).
Most invertebrates, fishes, amphibians, reptiles.	Birds and mammals.

13.3.2 Types of Animals on the basis of ability of Thermoregulation

Q.36 [Short] How do poikilotherms differ from homeotherms?

Poikilotherms	Homeotherms
Unable to maintain their body temperature within narrow limits.	Maintain a relatively constant internal body temperature.
Body temperature varies considerably and usually matches the environment.	Body temperature stays constant regardless of environmental fluctuations.
No effective internal “thermostat”.	Use a “thermostat” (hypothalamus in the brain) to trigger cooling or heating responses.
All invertebrates; fishes, amphibians and reptiles.	Mammals and birds, e.g., humans (about 37 °C).

Q.37 [Short] What advantage do poikilotherms have over homeotherms?

Poikilotherms do not need to consume massive amounts of food to “fuel” a constant body temperature. This allows them to survive in nutrient-poor environments where a homeotherm would starve.

13.3.3 Thermoregulation in Humans

Q.38 [Short] Why are humans called endothermic homeotherms?

Humans are **endothermic** because they generate most of their body heat through internal metabolism. They are **homeotherms** because they maintain their internal temperature at roughly 37 °C, whether they are in a very cold or a very hot environment. They regulate the body temperature through a negative feedback mechanism controlled by the hypothalamus.

Q.39 [Long] Explain how the human body regulates its temperature in hot and cold conditions.

Thermoreceptors present in the skin detect an increase or decrease in body temperature. They send signals to the control centre, i.e., the hypothalamus, which also contains thermoreceptors. The hypothalamus sends signals to muscles and glands (effectors), which produce corrective responses. This is a negative feedback mechanism.

Responses to overheating (hyperthermia) — temperature above 37 °C

- **Vasodilation:** It is the expansion of the blood capillaries which lie just beneath the epidermis of the skin. More blood flows in these capillaries, so more heat is removed from the blood.
- **Sweating:** Evaporation of sweat carries heat from the body and produces a cooling effect.
- **Decrease in thyroxine:** The thyroid gland may decrease the secretion of thyroxine, which lowers the basal metabolic rate. So the internal heat production is reduced.

Responses to cooling (hypothermia) — temperature below 37 °C

- **Vasoconstriction:** It reduces blood flow to the skin, so there is less loss of heat from the blood.
- **Shivering:** It is the spasmodic contraction of skeletal muscles. It produces heat, which helps to raise the body temperature.
- **Goosebumps:** Tiny muscles at the base of hair follicles contract. Air is trapped in the erected hair and acts as an insulator for heat.
- **Increased metabolic rate:** It also produces heat to raise the body temperature.

Effector	In overheating	In cooling
Skin capillaries	Vasodilation — more heat lost	Vasoconstriction — less heat lost
Sweat glands	Sweating — cooling by evaporation	—
Skeletal muscles	—	Shivering — heat produced
Hair muscles	—	Hair erected (goosebumps) — air trapped as insulator
Metabolic rate	Decreased (less thyroxine)	Increased

Q.40 [Short] How does the thyroid gland take part in thermoregulation?

Thyroxine, the hormone of the thyroid gland, controls the basal metabolic rate, and metabolism produces heat. In overheating, the thyroid gland may decrease the secretion of thyroxine. This lowers the basal metabolic rate and reduces internal heat production. In cooling, the metabolic rate is increased to produce more heat.

Q.41 [Short] Why does the skin look red when we feel hot and pale when we feel cold?

When we are hot, vasodilation occurs and more blood flows in the capillaries just beneath the epidermis, so the skin looks red and heat is lost. When we are cold, vasoconstriction reduces the blood flow to the skin, so the skin looks pale and heat is saved.

The colour change of the skin is an application of the textbook's vasodilation and vasoconstriction; it is not stated in the textbook.

Q.42 [Short] How do we "feel" warmth or coldness? What is the benefit of this feeling?

The thermoreceptors send signals not only to the hypothalamus but also stimulate the higher centres of the brain. So we "feel" warmth or coldness. This feeling allows us to take more actions (behavioural responses), like taking a bath or turning on the fans or heaters.

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